Student Fee Defaulter System

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Abstract: Timely monitoring of student fee payments is an important administrative task, particularly when institutions handle numerous student records and payment transactions. This paper presents a web-based Student Fee Defaulter Management System developed to maintain fee records, monitor outstanding amounts, identify defaulters, and automate payment-related communication. The system combines a Next.js and TypeScript interface with Java Spring Boot and Python Flask services, supported by a shared Supabase PostgreSQL database. Students can pay dues through Razorpay or submit bank-payment receipts for offline verification. Gemini 2.5 Flash is used to extract transaction details from uploaded receipts, which are subsequently reviewed by administrators. n8n and Brevo automate notifications, while completed payment details can be synchronized with Google Sheets. The system also incorporates OTP authentication, password validation, login-attempt control, and temporary account locking. By bringing fee tracking, payment processing, receipt verification, and notification management into one platform, the proposed system provides a centralized approach to handling student fee obligations.

Index Terms— Student Fee Management, Fee Defaulter System, Payment Monitoring, Automated Notifications, Receipt Verification, OCR, Online Payment, Workflow Automation**.**

I. INTRODUCTION

Fee administration is an important part of educational institutions as it involves maintaining student records, assigned fees, payments, and pending dues. Managing these activities through separate records can make fee tracking difficult and increase administrative effort. The proposed Student Fee Defaulter Management System brings these activities together in a single web-based platform.

The system allows administrators to manage student records, assign fees, monitor payments, identify overdue or

partially paid accounts, and send fee-related notifications. A centralized Supabase PostgreSQL database maintains student, fee, payment, and receipt information.
Students can pay fees online through Razorpay or submit bank receipts for offline payments. Gemini 2.5 Flash OCR extracts details such as UTR, amount, and payment date from uploaded receipts, which are then verified by the administrator.

The system uses n8n and Brevo for OTP verification, payment confirmations, overdue reminders, and challan-status notifications. Successful payments can also be logged in Google Sheets through an automated workflow.

The platform combines Next.js and TypeScript with Java Spring Boot and Python Flask services. Security features include OTP verification, password validation, login-attempt restrictions, and temporary account locking. Overall, the system provides a centralized approach to fee monitoring, payment processing, receipt verification, and defaulter management.

II. PROBLEM STATEMENT

Managing student fee records becomes difficult when assigned fees, payments, pending amounts, and due dates are maintained through separate processes. As the number of students increases, administrators may face difficulty in tracking individual payment status and identifying unpaid or partially paid fees.

Payment verification is another challenge because students may use both online and offline payment methods. Online payments require confirmation, while offline payments may require manual checking of bank receipts or challans. These processes can increase administrative effort and delay fee-record updates.
Communication also becomes repetitive when students require reminders for overdue or incomplete payments and notifications for successful payments or submitted receipts.

Therefore, a unified system is required to manage fee records, monitor outstanding payments, support online and offline payments, verify receipts, and automate notifications. The proposed Student Fee Defaulter Management System addresses these requirements through an integrated web-based platform.

III. LITERATURE REVIEW

Digital fee management provides educational institutions with structured methods for maintaining student, fee, and payment records. The proposed system follows this centralized approach to organize fee information and provide administrators with easier access to payment records.
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Monitoring unpaid and partially paid fees is an important part of fee administration. The system maintains information such as paid amount, remaining balance, due date, and payment status, allowing administrators to monitor overdue and partial payments through a common interface.

Online payment integration provides students with a convenient method for completing fee transactions. The system uses Razorpay for digital payments and connects verified transactions with the corresponding student fee records.

For offline payments, students can submit bank receipts or challans. Gemini 2.5 Flash OCR extracts details such as UTR, amount,

and payment date, which are then reviewed by the administrator before the fee record is updated.

Automated communication reduces repetitive fee-collection activities. The system uses n8n and Brevo for OTP verification, overdue reminders, partial-payment alerts, payment confirmations, and challan-status notifications. A scheduled process also supports reminders for overdue and partially paid accounts.

The system combines Next.js, TypeScript, Java Spring Boot, Python Flask, and Supabase PostgreSQL with external payment, OCR, email, and workflow services. Security mechanisms include OTP verification, password requirements, login-attempt restrictions, and temporary account locking, providing the technical foundation for the proposed fee management system.

V. SYSTEM ARCHITECTURE

The proposed Student Fee Defaulter Management System is organized as a layered architecture consisting of five major layers: the Client/User Layer, Application and Automation Layer, Central Database Layer, AI and Payment Gateway Layer, and Notification and Logging Pipeline. The separation of these layers allows different components of the system to perform their respective functions while remaining connected through the application services and central database.

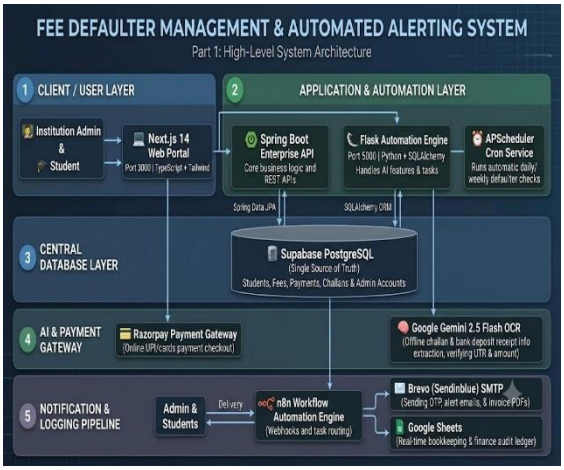


Fig1.System Architecture of the Proposed Student Fee Defaulter Management System

The Client/User Layer represents the users who interact with the system. It includes the institution administrator and students. Both users access the application through the Next.js 14 web portal, which is developed using TypeScript and Tailwind CSS. The portal provides the user interface required to access the features available to each type of user.

The Application and Automation Layer forms the main processing layer of the architecture. It contains the Spring Boot Enterprise API, Flask Automation Engine, and APScheduler Cron Service. The Spring Boot service provides the core REST APIs and handles the main application-level operations. The Flask service is responsible for automation-oriented and AI-related functions. APScheduler provides scheduled execution of predefined tasks, such as periodic fee and defaulter checks.

The Central Database Layer provides a common data storage area for the application. The system uses Supabase PostgreSQL as its central database. It stores the major records required by the system,

including student information, fee details, payment records, challan information, and administrator accounts. The database acts as the common storage layer accessed by the application services.

The AI and Payment Gateway Layer provides specialized external services required by the system. Razorpay Payment Gateway is integrated to support online fee payments through methods such as UPI and cards. Google Gemini 2.5 Flash OCR is used for processing uploaded offline challans and bank deposit receipts and extracting relevant payment information from them. These services extend the system with payment processing and AI-based document analysis capabilities.

The Notification and Logging Pipeline manages automated communication and additional payment-record logging. n8n Workflow Automation is used to connect events with the required automated tasks. Brevo SMTP supports the delivery of OTPs, fee alerts, and other email notifications. Google Sheets provides an additional location for recording payment-related information and maintaining a bookkeeping or audit record.

As shown in **Fig. 1**, these five layers form the overall technical structure of the proposed system. The architecture separates user interaction, application processing, data storage, specialized external services, and automated communication into distinct components, while the connected interfaces allow them to operate as a single fee management platform.

VI. WORKING OF THE PROPOSED SYSTEM

The proposed Student Fee Defaulter Management System provides separate interfaces for administrators and students and supports both online and offline fee payment methods. The overall operation consists of system setup, fee-status monitoring, student authentication, payment processing, receipt submission, verification, and notification. The complete sequence of these activities is illustrated in **Fig. 2.**

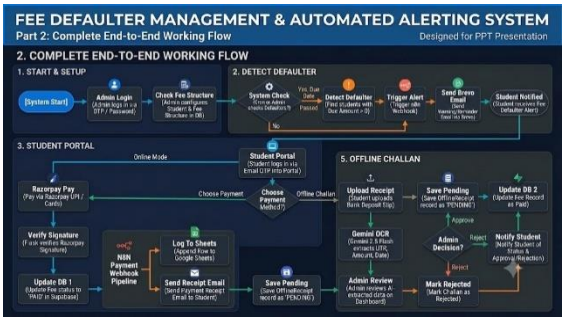


Fig 2.End to End Working flow

A. System Initialization and Fee Setup

The process begins with administrator authentication. After entering the required login credentials, the administrator accesses the fee management functions of the system. The administrator can configure the applicable fee structure and assign fee information to students. The stored fee details provide the basis for monitoring payment status and outstanding amounts.

B. Fee Monitoring and Defaulter Identification

The system continuously maintains the fee status of registered students. A scheduled checking mechanism examines the due information and identifies accounts having an outstanding amount after the applicable deadline. Students meeting the defined defaulter condition are included in the

defaulter records. The notification workflow can then generate an appropriate fee reminder and deliver it to the concerned student through email.

C. Student Authentication and Portal Access

Students access the system through the Student Portal using their registered roll number and email address. The authentication process uses OTP-based verification to provide controlled access to the student's fee information. After successful authentication, the student is directed to the personal dashboard.

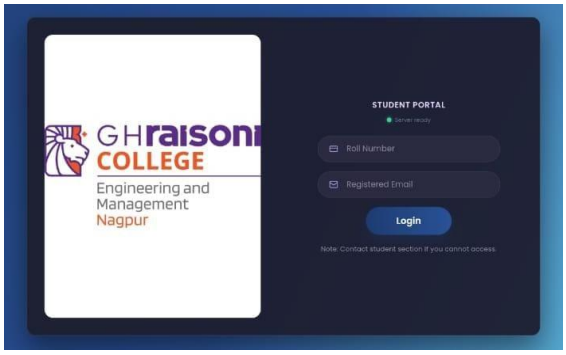


Fig 3. Student Portal Login Interface

The student dashboard displays important fee-related information, including the total course fee, amount already paid, applicable late fine, and remaining payable amount.

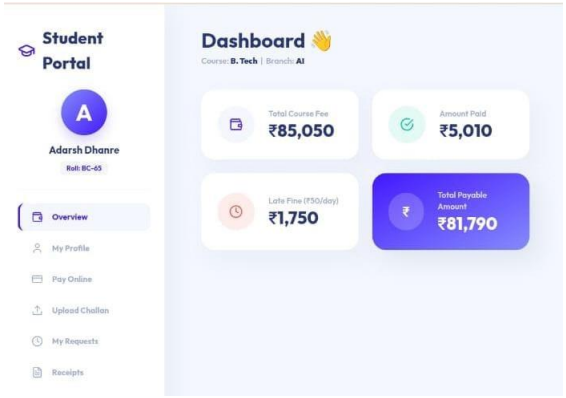


Fig 4. Student Fee Dashboard

D. Online Fee Payment

For digital payment, the student selects the online payment option from the portal. The system displays the payable amount and provides an option to initiate the transaction through the integrated payment gateway. After the payment is completed, the transaction information is verified before the corresponding fee record is updated.

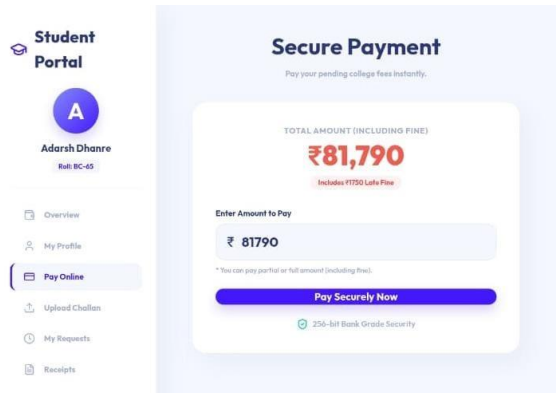


Fig 5. Online Fee Payment Interface

After successful verification, the payment is associated with the student's record. The payment workflow can also trigger a confirmation email and record the transaction information through the automated logging mechanism.

E. Offline Challan Submission and Verification

The system also supports students who complete their fee payment through an offline banking method. The student can upload a scanned or digital copy of the bank challan or payment receipt through the Student Portal.

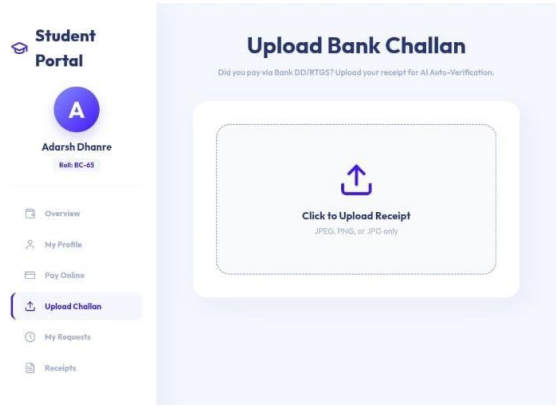


Fig 6. Offline Challan Upload Interface

F. Notification and Record Maintenance

The system uses workflow automation to manage communication related to authentication, fee reminders, successful payments, and challan status. Payment information is also maintained in the central database, while successful transactions can be logged through the connected Google Sheets workflow. This allows student fee information and payment records to remain organized throughout the payment cycle.

The complete working process therefore provides two payment paths: an online route through the payment gateway and an offline route through challan submission and administrative verification. Both routes are connected to the student's fee record and the administrative monitoring system.

VII. RESULTS AND DISCUSSION

The developed Student Fee Defaulter Management System successfully integrates student fee management, payment monitoring, defaulter identification, and administrative operations into a single web-based platform.

The Admin Dashboard provides an overall view of student fee status, including fully paid, partially paid, and overdue accounts. It also displays pending fine information, fee collection status, and quick actions for managing payments and defaulters.

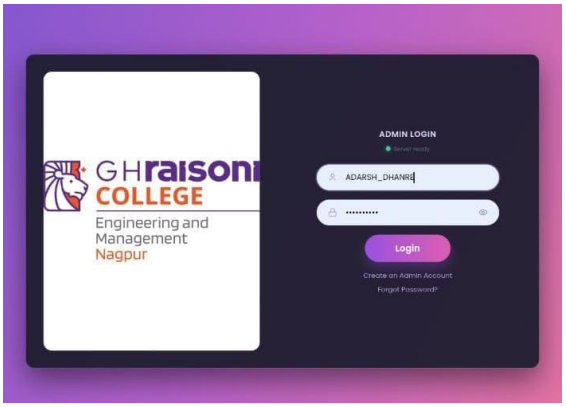


Fig 7.Administrator Dashboard

The Defaulter Monitoring Interface provides administrators with a structured view of students having outstanding fees. It displays the total fee, amount paid, outstanding balance, and due information, allowing pending accounts to be monitored and fee reminders to be initiated**.**

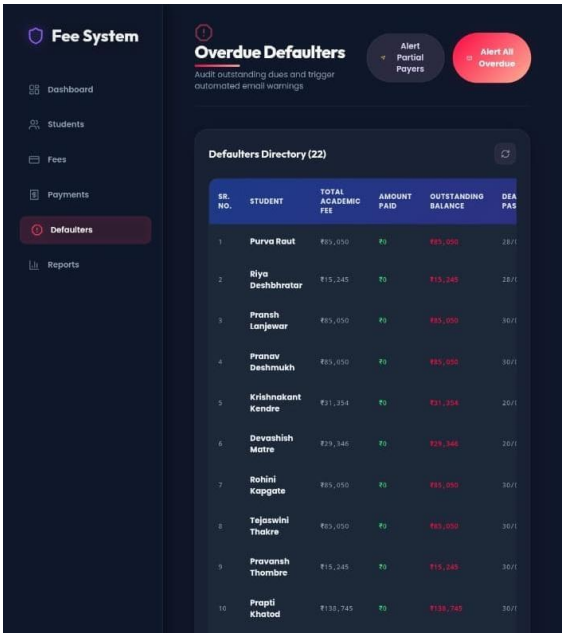


Fig 8. Fee Defaulter Monitoring Interface

The Fee Administration Interface allows administrators to manage student fee

allocations and review payment-related information. The interface displays assigned fees, paid amounts, remaining dues, and applicable late fines, providing a centralized view of individual fee records.

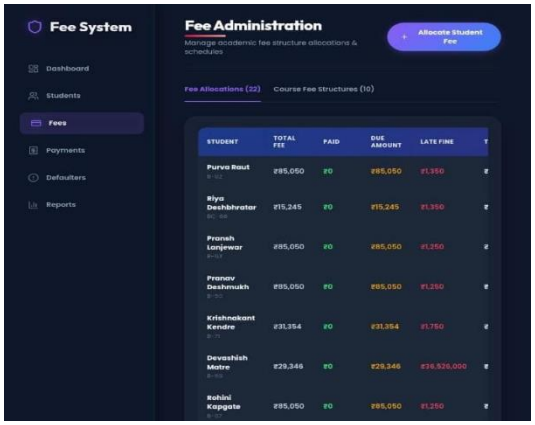


Fig. 9. Fee Administration and Allocation Interface

Overall, the implemented system demonstrates a connected approach to student fee administration.. The developed prototype therefore provides a structured platform for managing the student fee-payment cycle and monitoring outstanding dues.

VII. FUTURE SCOPE

The proposed Student Fee Defaulter Management System can be extended further to support larger institutional requirements. Future development may include a dedicated mobile application so that students and administrators can access fee-related services more conveniently. The system can also be enhanced with additional payment gateways and more payment options to provide greater flexibility.

The receipt-verification module can be improved to handle different receipt formats and improve extraction accuracy for unclear or low-quality documents. More detailed analytics and graphical reports could also be added to help administrators study payment patterns, outstanding dues, and fee collection trends.

Future versions may support integration with existing college ERP or student information systems to reduce duplicate data entry. Notification support could also be expanded to additional communication channels such as SMS or other institutional messaging services. With these improvements, the system could be adapted for larger institutions and provide more comprehensive automation for student fee administration.

VIII. CONCLUSION

The Student Fee Defaulter Management System provides a centralized solution for handling student fee-related activities. The system connects the Student Portal and Admin Panel, allowing students to check their fee status, view payment details, make online payments, and submit offline payment receipts. At the same time, administrators can manage student records, assign fees, monitor pending payments, and identify students with overdue or partially paid fees.

The integration of Razorpay, Gemini 2.5 Flash, n8n, Brevo, and Google Sheets extends the functionality of the system beyond basic fee management. Online payments can be processed through Razorpay, while uploaded bank receipts can be analyzed to extract transaction details for administrator verification. Automated workflows support OTP verification, payment confirmations, overdue reminders, and other fee-related communication. These features help connect different stages of the fee-payment process within one platform.

Overall, the developed prototype demonstrates a practical approach to

organizing student fee administration through centralized records and automation. Security mechanisms such as OTP authentication, password validation, login-attempt restrictions, and temporary account locking are also incorporated. The system provides a structured workflow for students and administrators and establishes a foundation that can be further extended with additional features and institutional integrations.

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